

WEAVE: Overcoming the Identity Shortcut in Style Transfer via Minute-Scale Wavelet-Decoupled Flow Matching

Anonymous submission

Abstract

Current style transfer methods face two primary challenges. First, state-of-the-art diffusion models, along with training-free methods built on them, require more than 60 GB of VRAM and therefore cannot run on consumer GPUs. Second, lightweight methods without these generative backbones often underperform a trivial identity mapping, which scores high on traditional metrics like ArtFID and LPIPS but was rarely evaluated as a baseline.

We empirically observe that frequency dominance in shared representation spaces pushes lightweight models toward identity-like solutions (Rahaman et al. 2019): low-frequency structural energy overwhelms high-frequency style gradients, a pattern we confirm through gradient-spectrum analysis and per-subband style-transfer measurements.

We argue that an effective style transfer method should outperform the identity baseline while remaining computationally affordable. To this end, we propose WEAVE: a single-level Haar

wavelet transform splits the VAE latent into a low-frequency structure band and high-frequency style bands; a compact rectified-flow network learns a structure-preserving velocity field; and endpoint statistical alignment (AdaIN) injects the target style into the high-frequency subbands. A fixed-latent gradient probe stops training when high-frequency optimization begins to dominate the shared trunk, without decoding images or consulting evaluation metrics. This decoupling lets WEAVE train in under two minutes on a single consumer GPU.

We also advocate the IDT–TGT sandwich as an intuitive evaluation guideline. IDT returns the source unchanged, while TGT returns an unrelated image from the target style; together they show whether an output gains style without giving up source content beyond the target-reference level. Under this guideline, WEAVE clears the identity reference with 1.04M trainable parameters (plus a frozen 84M VAE), making high-quality style transfer both effective and affordable.

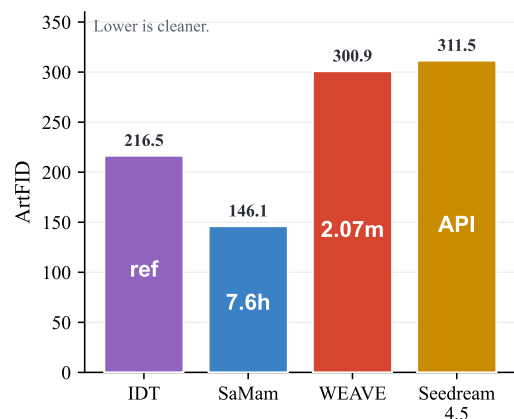
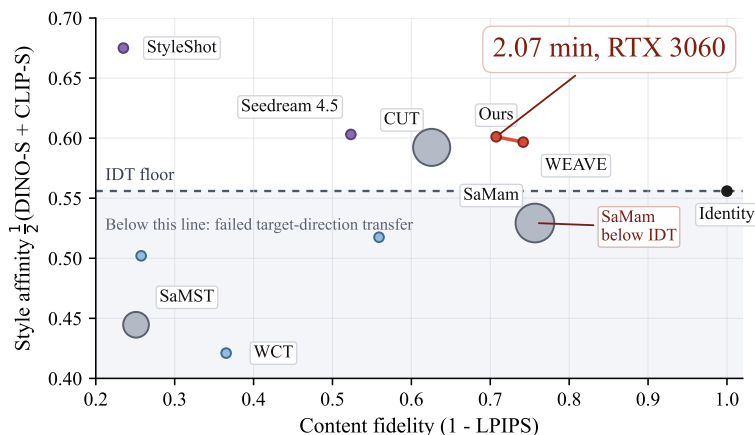


Figure 1: IDT–TGT sandwich comparison on Distinct5-WikiArt. Left: style affinity $\frac{1}{2}(\text{DINO-S} + \text{CLIP-S})$ versus $1 - \text{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned). The dashed line marks the IDT style reference (0.556); methods below it have not moved in the requested direction. The dotted line marks the TGT content reference (0.267); methods right of it are perceptually farther from the source than an unrelated target-style image. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

Introduction

Style transfer faces two linked problems in deployment and evaluation. Large diffusion editors and training-free methods

built on them can require more than 60 GB of GPU memory, which places their strongest priors outside consumer hardware. Lightweight alternatives avoid this cost, but some preserve the source so strongly that their style affinity falls below the unchanged input. Conventional protocols can hide this no-op behavior: LPIPS and ArtFID strongly reward structural reconstruction, yet the identity mapping is rarely reported as an explicit baseline.

Figure 1 makes the missing comparison visible with two simple outputs. Identity (IDT) returns the source unchanged, while Target Style (TGT) returns an unrelated reference from the requested style. A useful transfer should move beyond IDT in the style direction without moving as far from the source as TGT in the content direction. The two references expose opposite failures in the original metric units: methods below IDT have not acquired enough target style, whereas methods beyond TGT have discarded at least as much source content as a source-independent style image. We call this the IDT–TGT sandwich. It is an empirical diagnostic, not a replacement for DINO, CLIP, LPIPS, or visual inspection.

Why do compact models enter the identity regime? We argue that an important driver is the shared coordinate system rather than parameter count alone. Neural networks learn low frequencies first (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), and natural images place much more energy in global layout and object shape than in brushstroke and fine texture. A flow-matching model trained on the full latent must explain both with one velocity field. The resulting low-frequency residuals are larger and easier to fit, so their gradients dominate the update and weaken style-conditioned high-frequency motion. Frequency-domain translation reports a related content–style conflict (Cai et al. 2021); our gradient and subband probes measure it directly in the trained model. Simply increasing width does not change this competition because both signals still share the same basis and objective.

We therefore propose WEAVE, which changes the representation and target before increasing capacity. A frozen diffusion VAE supplies a compact latent, while a one-level Haar transform separates low-frequency structure (LL) from oriented texture (LH/HL/HH). A 1.04M-parameter rectified-flow model learns source-aligned transport with weaker LL supervision; compact orientation-specific target codes modulate the matching LH/HL heads; and one endpoint AdaIN update supplies target-style statistics to the high-frequency bands. Cached latents remove the VAE encoder and diffusion U-Net from training; eight compact Euler evaluations and one VAE decode are sufficient at inference. An internal gradient transition selects the fourth epoch without image evaluation, giving a 1.4-minute run on one RTX 3060; the same endpoint operator can be added to frozen SD1.5 without retraining (Section).

Contributions. (1) **Evaluation:** We advocate the IDT–TGT sandwich, which reports a no-op reference and a source-independent target-style reference alongside standard metrics, exposing both style suppression and content collapse. (2) **Representation and method:** We diagnose low-frequency gradient dominance in a shared latent objective and develop a wavelet-decoupled flow that assigns structure, learned texture

transport, and endpoint statistics to explicit paths. (3) **Efficiency and evidence:** WEAVE uses 1.04M trainable parameters and 1.4 minutes of single-GPU training. Cross-dataset evaluation, internal gradient probes, and a frozen SD1.5 plugin test generalization and portability.

Related Work

Neural Style Transfer

Artistic style transfer was introduced by optimizing pixels to match content features and Gram statistics of a pretrained network (Gatys, Ecker, and Bethge 2015, 2016). AdaIN aligns channel-wise moments (Huang and Belongie 2017), while WCT aligns a full feature covariance (Li et al. 2017); both replace iterative optimization with a feed-forward feature transform. SANet and transformer-based variants add correspondence or long-range attention (Park and Lee 2019; Deng et al. 2022; Hong et al. 2023; Kwon et al. 2024), and CUT learns unpaired translation through patch-level contrast (Park et al. 2020). Training-free methods also reuse pretrained codebooks or hierarchical vision transformers (Gim, Yoo, and Uh 2024; Zhang et al. 2024a). The survey of Zhang et al. (Zhang and Tang 2025) summarizes this progression. These methods differ in supervision and backbone, but most still mix structure and appearance in one feature tensor. Stronger moment or attention transfer can therefore alter source geometry; WEAVE instead restricts the analytic style correction to explicit frequency coordinates.

Diffusion-based and Training-free Style Editing

Latent diffusion models (Rombach et al. 2022) move editing into a compressed generative space and provide strong semantic priors. Prompt-to-Prompt and SDEdit control generation through attention or noising (Hertz et al. 2022; Meng et al. 2021); StyleID and StyleAligned modify attention statistics (Chung, Hyun, and Heo 2024; Hertz et al. 2024); IP-Adapter adds an image prompt (Ye et al. 2023); and InSt, DiffuseIT, ArtBank, and Z-STAR use inversion or pretrained features for style transfer (Zhang et al. 2023; Kim, Kwon, and Ye 2023; Zhang et al. 2024b; Lu et al. 2024). These methods are often called training-free because they do not update backbone weights, but inference still invokes a large denoiser repeatedly and may require inversion. Recent FLUX and Qwen-Image backbones further increase the memory cost (Black Forest Labs 2024; Alibaba Cloud 2024). WEAVE retains only the frozen VAE and learns the remaining transport with a compact model; the SD1.5 plugin separately tests whether its analytic endpoint operation transfers back to a large pretrained editor.

Flow Matching and Schrödinger Bridges

Rectified flow (Liu, Gong, and Liu 2023) and flow matching (Lipman et al. 2022) regress a velocity along an interpolation between endpoint distributions, enabling short numerical paths. Schrödinger bridges (Léonard 2014) add entropy-regularized stochastic transport, and DSB, DSBM, I2SB, and DDIB develop practical diffusion or flow solvers for unpaired domains (De Bortoli et al. 2021; Shi et al. 2023; Liu et al. 2023; Su et al. 2023). These methods are related

to regularized optimal transport (Cuturi 2013; Peyré and Cuturi 2019), but their image objectives usually measure all coordinates together. WEAVE does not replace the rectified-flow path. It constructs a source-aligned endpoint, applies the same path in an orthogonal wavelet basis, and weights its blocks separately. This preserves the few-step advantage while changing which errors compete during learning.

Lightweight Models and Evaluation

Mamba-based SaMST and SaMam reduce trainable size and sequence cost (Liu et al. 2024, 2025; Gu and Dao 2023), showing that style transfer need not inherit a full diffusion U-Net. Efficiency alone, however, does not establish that the output moved toward the requested style. LPIPS measures perceptual source distance (Zhang et al. 2018), and ArtFID combines style distribution and content terms (Wright and Ommer 2022); both can reward a near-copy when the identity output is omitted. IDT makes this no-op score explicit. TGT adds the opposite reference: a target-style image with no source dependence. Reporting both rows preserves the standard metrics and their raw units, while identifying whether an apparent gain comes from style suppression or content collapse. The references are benchmark-specific diagnostics rather than universal thresholds.

Spectral Bias and Frequency-aware Optimization

Neural networks preferentially fit low-frequency functions (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), and natural images place most spectral energy in coarse structure. The two effects are distinct but reinforce each other: the training target supplies larger low-frequency residuals, while the network fits those residuals earlier. Frequency-domain translation therefore constrains low and high frequencies separately to protect content identity (Cai et al. 2021). Our contribution is to connect this optimization imbalance to the identity shortcut in a compact flow model and measure the connection through per-band loss, input gradients, head gradients, and endpoint interventions. The measurements motivate a structural preconditioner rather than another scalar style weight.

Wavelet and Latent-space Alignment

Wavelets separate local information by scale and orientation and support exact reconstruction. They have been used in restoration and generation (Phung, Dao, and Tran 2023) and in photorealistic style transfer (Yoo et al. 2019). The Haar basis is particularly simple: its orthogonality gives additive energy decomposition, and its compact support preserves local correspondence (Daubechies 1992). VAE latents provide a complementary reduction in spatial cost (Rombach et al. 2022), while AdaIN and WCT provide closed-form statistical alignment. WEAVE combines these ingredients inside the transport objective rather than using wavelets only as preprocessing or postprocessing. LL, LH, and HL receive explicit velocity roles; HH is reserved for endpoint statistics. This division is the central difference from a single latent-space transform such as Latent-WCT.

Method

A direct content-to-style target changes layout, palette, and texture at once. Under shared latent MSE, the dominant structural residual sets the update, so increasing style strength often spends content instead of resolving the conflict. WEAVE instead constructs a source-aligned endpoint, learns transport in orthogonal Haar bands, and applies one high-frequency statistics update after the path (Figure 2).

Why a Shared Latent Objective Favors Structure

Consider a velocity error $e_\theta = v_\theta(z_t, s) - u$. Because the Haar transform is orthonormal, an unweighted latent MSE can be written exactly as

$$\mathcal{L}_{\text{shared}} = \mathbb{E}_t \|e_\theta\|_2^2 = \mathbb{E}_t \left[\|e_\ell\|_2^2 + \sum_{i=1}^3 \|e_{h_i}\|_2^2 \right]. \quad (1)$$

The sum alone is not harmful; the problem is its scale and shared parameterization. Let $J_b = \partial v_b / \partial \theta$ be the output Jacobian of band b . The parameter update is

$$\nabla_\theta \mathcal{L}_{\text{shared}} = 2 \mathbb{E}_t \left[J_\ell^\top e_\ell + \sum_{i=1}^3 J_{h_i}^\top e_{h_i} \right]. \quad (2)$$

Natural-image geometry makes $\|e_\ell\|$ large, while spectral bias makes low-frequency functions easier to fit (Rahaman et al. 2019; Xu, Zhang, and Luo 2022). Unless the high-frequency Jacobians compensate, the LL term sets the early descent direction and the model underuses style conditioning. The measured parameter gradients show the same imbalance (Section). WEAVE changes the endpoint residual, separates output heads, and downweights LL before gradients enter the shared backbone.

Framework Setup

Let $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times 32 \times 32}$ and $z_s = \mathcal{E}(x_s)$ be content and target-style latents from the frozen SD1.5 VAE (Rombach et al. 2022). A direct path from z_c to z_s also learns the unrelated geometry of x_s . We instead decompose $\mathcal{W}(z_c) = (\ell_c, h_{1,c}, h_{2,c}, h_{3,c})$ and similarly for z_s , then form the structure-aligned endpoint

$$\begin{aligned} \tilde{\ell}_s &= (1 - \alpha)\ell_c + \alpha \text{AdaIN}(\ell_c, \ell_s), & \alpha &= 0.3, \\ z_1^* &= \mathcal{W}^{-1}(\tilde{\ell}_s, h_{1,s}, h_{2,s}, h_{3,s}). \end{aligned} \quad (3)$$

The LL blend permits coarse appearance to move while keeping source layout; the high-frequency target comes from the requested style. Exact source LL suppresses palette change, whereas the complete style latent would also supervise unrelated target geometry.

The corresponding target velocity is explicit in wavelet coordinates. Writing $\mathcal{W}(z_1^* - z_c) = (u_\ell, u_{h_1}, u_{h_2}, u_{h_3})$,

$$\begin{aligned} u_\ell &= \alpha[\text{AdaIN}(\ell_c, \ell_s) - \ell_c], \\ u_{h_i} &= h_{i,s} - h_{i,c}, & i &\in \{1, 2, 3\}. \end{aligned} \quad (4)$$

Thus α bounds the supervised coarse movement, while the high-frequency residuals retain the target-style signal. This is different from merely reducing the LL loss weight: Eq. 4

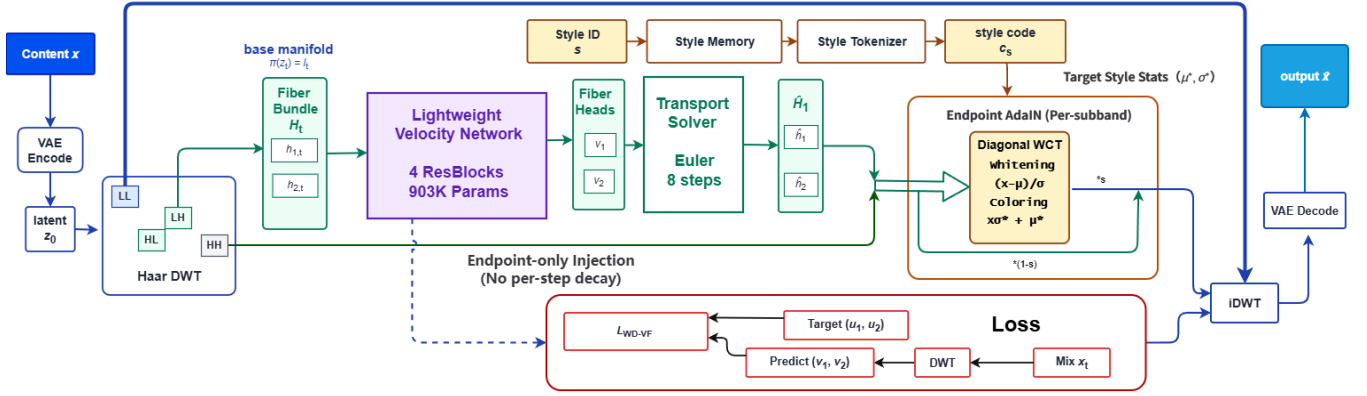


Figure 2: **WEAVE architecture.** A single-level Haar transform exposes the LL/LH/HL/HH subbands of the SD1.5 VAE latent. The compact base velocity network predicts LL/LH/HL motion, while pooled target LH/HL codes gate residuals into their matching heads; HH carries no learned head. Band-weighted rectified-flow matching learns the transport, and per-subband AdaIN aligns LH/HL/HH once at the endpoint. The complete D5 model has 1.04M trainable parameters.

changes what the model is asked to predict, whereas the weight controls how strongly an error changes the parameters.

Rectified flow learns the straight path

$$z_t = (1-t)z_c + tz_1^*, \quad u = z_1^* - z_c, qquad t \sim \mathcal{U}([0, 1]). \quad (5)$$

Because $\mathcal{W}$ is linear, the same interpolation holds independently in every subband. WEAVE therefore keeps the standard rectified-flow target and changes only the endpoint, parameter sharing, and loss weights.

Wavelet Decomposition and Frequency Routing

Low-frequency structure carries much more energy than fine texture, so a shared loss favors source reconstruction (Rahaman et al. 2019; Xu, Zhang, and Luo 2022; Cai et al. 2021). WEAVE separates these signals before predicting the flow.

Haar wavelet decomposition. A one-level Haar transform gives $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$: one LL structure band and three high-frequency bands, each $4 \times 16 \times 16$. Haar is orthogonal and local, so high-frequency updates do not directly alter the LL coordinate (Daubechies 1992).

$$\|\Delta z\|_F^2 = \|\Delta \ell\|_F^2 + \sum_{i=1}^3 \|\Delta h_i\|_F^2. \quad (6)$$

Equation 6 is simply Parseval’s identity. It makes the design transparent: an operation restricted to LH/HL/HH leaves the LL coefficient unchanged, although the nonlinear VAE decoder can still couple visible color and texture.

Separate 1×1 heads then remove output-level interference even though the backbone is shared: the aligned endpoint defines the movement, λ_{LL} controls structural gradient pressure, and AdaIN handles statistics that spatial MSE learns poorly.

Empirical frequency probes. Across 5,000 cross-style latent pairs, transport energy concentrates in LL while style separation is strongest in HH (Figure 3). This motivates a weak LL loss and stronger high-frequency routing.

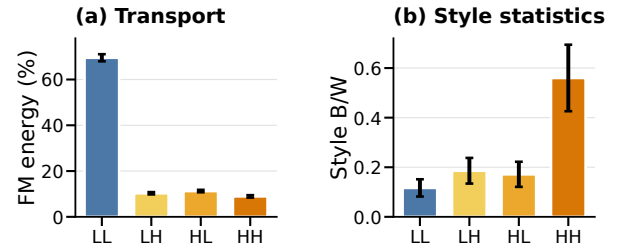


Figure 3: **Frequency probes.** (a) Cross-style transport energy in each Haar band. (b) Between-style versus within-style separation of latent statistics. Error bars summarize variation across style routes or statistics.

Architecture. The base velocity network has four width-64 residual blocks and separate 1×1 heads for LL, LH, and HL. HH has no velocity head and changes through the endpoint operation. Learned style-memory tokens provide a category-level prior to the shared backbone; the ablations show that this branch is auxiliary rather than the main style-injection path. For target-specific detail, convolution–pool encoders produce LH/HL codes c_i and update only the matching head as $v_{h_i} = v_{h_i}^{\text{base}} + \tanh(g_i)R_i(H, c_i)$. Pooling removes target coordinates, so the 1.04M-parameter model carries oriented appearance without copying reference layout into LL. HH is deliberately statistical rather than transported. Although it carries stroke and grain statistics, its phase is not registered in unpaired transfer: $h_{3,s}(i, j)$ is not the desired detail for source location (i, j) . A pointwise HH head would minimize $\|v_{h_i} - (h_{3,s} - h_{3,c})\|_2^2$, whose optimum averages incompatible phases and sends noisy gradients through the shared backbone. Endpoint AdaIN instead matches per-sample HH moments without learning target-image coordinates; a matched HH-head ablation did not improve the style-content frontier.

Why the model is fast. For a 512×512 image, the frozen VAE produces only 32×32 spatial positions; the velocity

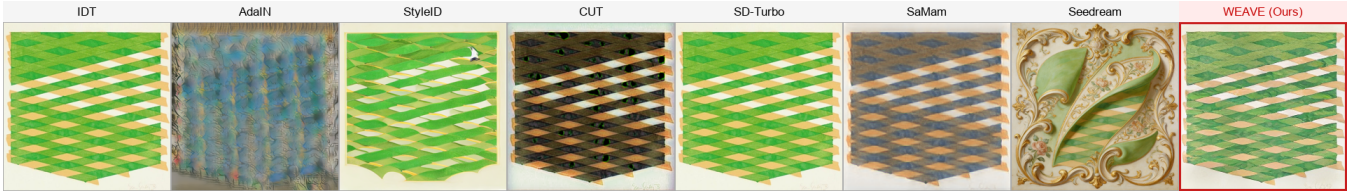


Figure 4: Extreme cross-domain stylization (Minimalism → Rococo). Diffusion editors substantially alter source structure, while several lightweight or classical outputs remain close to identity. WEAVE preserves source topology through low-frequency anchoring and injects painterly texture through high-frequency subbands.

backbone operates on 16×16 subbands, versus 512×512 RGB positions. Cached latents remove the encoder and the large diffusion U-Net from training. Rectified flow supplies target velocities directly, so there is no adversarial loop, denoising teacher, or inversion stage; inference uses a short path, one analytic transform, and one VAE decode.

With C_v for one velocity evaluation and C_{dec} for one frozen VAE decode,

$$C_{\text{WEAVE}} \approx KC_v + C_{\text{AdaIN}} + C_{dec}. \quad (7)$$

Here C_v acts on small subbands and C_{AdaIN} is a linear-time moment update, whereas training-free diffusion editors still repeatedly evaluate a large denoiser.

Spectral flow-matching objective. The spectral flow-matching term is

$$\mathcal{L}_{\text{SFM}} := \mathbb{E}_t [\lambda_{LL} \|v_\ell - u_\ell\|_2^2 + \|v_{h_1} - u_{h_1}\|_2^2 + \|v_{h_2} - u_{h_2}\|_2^2], \quad (8)$$

with $\lambda_{LL} = 0.3$. The lower LL weight limits structural dominance while still allowing coarse tone to move. The predicted endpoint is $\hat{z}_1 = z_c + \mathcal{W}^{-1}(v_\ell, v_{h_1}, v_{h_2}, 0)$.

Equation 8 is a diagonal preconditioner. With $D = \text{diag}(\lambda_{LL}I, I, I, 0)$ and $e = \mathcal{W}(v - u)$, its output-space gradient is $2De$. The relative LL pressure before the shared Jacobian is therefore

$$r_{\text{out}} = \frac{\lambda_{LL} \|e_\ell\|_2}{(\|e_{h_1}\|_2^2 + \|e_{h_2}\|_2^2)^{1/2}}. \quad (9)$$

Choosing $\lambda_{LL} = 0.3$ reduces this pressure without deleting the coarse target. Parameter gradients are not perfectly separated because the backbone is shared, which is why Section measures them directly. HH is omitted from the velocity loss because the network has no HH head; its style statistics are supplied at the endpoint.

Internal-dynamics checkpoint selection. Endpoint loss may improve after perceptual style declines. We therefore probe a fixed latent minibatch at $t = 0.5$ and, for shared-trunk parameters θ_s , define

$$\rho_e = \frac{\|\nabla_{\theta_s} \mathcal{L}_{LL}\|_2}{\|\nabla_{\theta_s} (\mathcal{L}_{LH} + \mathcal{L}_{HL})\|_2 + \epsilon}. \quad (10)$$

With $\bar{g}_e = \frac{1}{2}[\tanh(g_1) + \tanh(g_2)]$, we stop at the first $e \geq 3$ where $\Delta \bar{g}_e > 0$ and ρ_e crosses below one: the target-HF route is expanding as HF gradients overtake LL. The probe uses two latent-space backward passes and no image or external metric; it selects epoch four (Section).

Endpoint AdaIN: Primary Style Injection Channel

Flow matching alone carries a weak style signal. WEAVE therefore applies AdaIN once, after the last Euler step, to align high-frequency statistics without repeatedly disturbing the transport path. Removing this endpoint update largely returns the model to source preservation (Table 2).

Alignment operator. Let $\hat{h}_i$ and h_i^* be high-frequency subbands of the predicted latent and a target-style reference. For $i \in \{1, 2, 3\}$, AdaIN (Huang and Belongie 2017) gives

$$T_i(a) = \frac{a - \hat{\mu}_i}{\hat{\sigma}_i} \sigma_i^* + \mu_i^*, \quad (11)$$

where μ and σ are channel-wise spatial statistics. We blend this update with scales $(s_1, s_2, s_3) = (0.3, 0.3, 0.5)$:

$$\hat{h}_i^+ = (1 - s_i) \hat{h}_i + s_i T_i(\hat{h}_i), \quad i \in \{1, 2, 3\}, \quad (12)$$

LL is left unchanged in this endpoint operation.

For positive channel standard deviations, the update has a direct statistical meaning:

$$\begin{aligned} \mu(\hat{h}_i^+) &= (1 - s_i) \hat{\mu}_i + s_i \mu_i^*, \\ \sigma(\hat{h}_i^+) &= (1 - s_i) \hat{\sigma}_i + s_i \sigma_i^*. \end{aligned} \quad (13)$$

The scale s_i therefore specifies how far each moment moves toward the reference, without requiring the velocity network to infer this global correction from spatial MSE. Applying the map only after the final Euler step keeps the correction out of later velocity updates. LL anchoring protects layout, the learned LH/HL path carries organized texture, and Table 1 shows that endpoint statistics alone are insufficient.

Training and Inference

Training adds a light edge ℓ_1 term and low-pass anchor to Eq. 8; together they contribute less than 5% of measured gradient mass. At inference, $z^0 = z_c$ and, for $K = 8$,

$$z^{k+1} = z^k + \frac{1}{K} \mathcal{W}^{-1}(v_\ell^k, v_{h_1}^k, v_{h_2}^k, 0), \quad (14)$$

then apply high-frequency AdaIN once and decode with the frozen VAE. Full architecture, training fields, and step ablations are in the supplement.

Experiments

Setup and Protocol

Datasets. Distinct5-WikiArt-512 (D5) contains five deliberately separated styles, with 3,600 training and 150 test images

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$	M	min	750 imgs
Identity	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
Target Style	1.000	0.874	0.733	0.176	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s
SD-Turbo	0.484	0.693 $\dagger$	0.003	0.922	0.341 $\dagger$	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0 $^{+1100}$	free	5 m
StyleAligned	0.675	0.780	0.869 $\ddagger$	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829 $\ddagger$	0.315	0 $^{+1100}$	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.514	0.786	0.332	0.526	0.514	<u>0.822</u>	0.384	0.526	0 $^{+1100}$	free	3 h
StyleShot	<u>0.563</u>	<u>0.787</u>	0.765 $\ddagger$	0.377	<u>0.552</u>	<u>0.774</u>	0.713	0.335	0.484	0.787	0.795 $\ddagger$	0.380	0 $^{+2400}$	free	5.1 h
StyleID	0.548	0.822	0.552	0.416	0.256 $\dagger$	0.648 $\dagger$	0.691	0.147 $\ddagger$	<u>0.571</u>	0.815	0.493	0.374	0 $^{+1100}$	free	63 m
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710 $\dagger$	0.620	<u>0.795</u>	7.0	322.6	5 m
SaMST	0.271 $\dagger$	0.618 $\dagger$	0.749 $\ddagger$	0.145 $\ddagger$	0.501	0.709	0.398	0.838	0.461	0.667 $\dagger$	0.612	0.615	8.3	39.5	10 m
SaMam	0.477	0.582 $\dagger$	<u>0.243</u>	<u>0.812</u>	0.505	0.677	0.205	0.870	0.503	0.712 $\dagger$	0.227	0.925	8.8	436.0	17.6 m
StyTR-2	0.501	0.709	0.596	0.747	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35.4 $^{+12.9}$	—	4 m
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11.3 $^{+20.0}$	—	3 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	<u>0.227</u>	0.785	0.504	0.742	0.486	0.725	—	API	—
Latent-WCT	0.362 $\dagger$	0.673 $\dagger$	0.441	0.559	0.338 $\dagger$	0.738	0.585	0.386	0.414 $\dagger$	0.710 $\dagger$	0.444	0.553	0	free	18 s
Ours	0.4915	0.7126	0.2596	0.8103	0.4801	0.6681	0.3116	<u>0.8612</u>	0.5226	0.7747	<u>0.2895</u>	0.7717	1.04 $^{+84}$	1.4	106 s

Table 1: Main comparison. DINO-S, CLIP-S, and DINO-C are higher better; LPIPS is lower better. Among methods, excluding the IDT/TGT references, the best and second-best quality values are bold and underlined. $\dagger$: style does not exceed IDT. $\ddagger$: content does not remain inside TGT. The D5 row uses the oriented-HF route and internal-dynamics epoch selection; P2A and R5 use the base checkpoints. Params are in millions; superscripts denote frozen parameters. Train is wall time on one RTX 3060; Infer is generation-only time for 750 images.

per style; its $5\times 5\times 30$ board has 750 requests. P2A-256 tests lower resolution, and R5 expands to all 20 WikiArt families (12,000 requests). DINO-S, our primary style metric, compares output DINOv2 features with the target-style pool; CLIP-S is secondary. DINO-C and LPIPS measure source retention. Table 1 compares twelve representative baselines under all four views.

IDT copies the source and TGT returns a fixed source-independent target-style image. Table symbols mark values outside the resulting bracket for that board and metric, not global method rankings. Validation uses 50 held-out pairs; boards, reference pools, preprocessing, and seeds are fixed. Full construction and hyperparameters are in the supplement.

Main Results

Reading the sandwich. The marks expose methods that remain below IDT on style or cross TGT on content distance. The bracket is board- and metric-specific: DINO-S/CLIP-S capture different style cues, while LPIPS/DINO-C capture different content failures. We therefore report each raw metric and compute IDT/TGT per board rather than collapsing them into one score.

D5 result. WEAVE clears IDT on both style measures while retaining substantially more source content than editors with a similar or stronger style score. Among trained methods that clear the CLIP-S identity reference, it has the lowest LPIPS and uses only 1.04M trainable parameters. The oriented target-HF route raises both DINO-S and CLIP-S over the base point without crossing the TGT content reference. The comparison with Seedream is especially informative: WEAVE reaches higher DINO-S with substantially lower content cost.

P2A and R5. WEAVE remains inside the bracket on P2A; on R5 it reaches the strongest DINO-S among trained

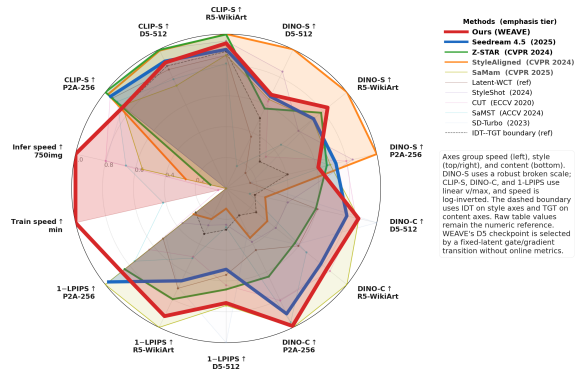


Figure 5: Style, content, and cost across three benchmarks. The dashed boundary uses IDT on style axes and TGT on content axes. DINO-S uses a broken visual scale; other metric axes are linearly normalized and speed axes are log-inverted. The D5 point uses the oriented-HF route selected by internal dynamics; Table 1 gives raw values.

lightweight methods while retaining low source distance. The stronger D5-512 profile is consistent with native high-frequency coefficients preserving detail compressed at 256. Because D5 and P2A also differ in data, we treat this as scale potential rather than an isolated resolution claim. Figure 5 summarizes the reported operating points.

Cost. WEAVE is substantially cheaper to train and run than the learned baselines in Table 1. Latent-WCT is faster still but falls below IDT, showing that analytic wavelet statistics alone do not replace learned transport.

Mechanistic Ablations

Table 2 compares matched diagnostic variants on D5. Removing AdaIN or flow matching shifts the result toward source preservation, while removing cross-attention changes every metric by at most 0.010. One Euler step and strong extrapolation degrade both objectives, identifying the active paths and their stable operating range.

Config	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-S $\uparrow$	DINO-C $\uparrow$
Baseline	0.727	0.343	0.483	0.755
w/o AdaIN	0.708	0.334	0.484	0.816
w/o Flow Matching	0.710	0.296	0.475	0.819
w/o Cross-attention	0.726	0.336	0.486	0.765
$\lambda_{LL}=0$	0.713	0.313	0.480	0.804
$lr = 5 \times 10^{-4}$	0.730	0.356	0.487	0.738
AdaIN $\alpha=0$	0.709	0.339	0.483	0.811
AdaIN $\alpha=2.0$	0.717	0.297	0.485	0.802
extrap = 1.0	0.704	0.591	0.411	0.616
num_steps = 1	0.709	0.476	0.376	0.448

Table 2: Ablation on D5 (750 pairs). The reported D5 model uses a 0.3 LL target blend, the oriented target-HF route, and the internally selected fourth epoch. Changes below 0.002 CLIP-S are omitted.

Setting $\lambda_{LL} = 0$ moves the model toward the source, so weak LL supervision is useful for coarse tone. Additional diagnostic sweeps are reported in the supplement.

Generalization and Robustness

Representation and adaptation. A matched RGB control performs worse on both style and content. Updating only the style memory from 30 images clears IDT on eight held-out styles, with little gain from full fine-tuning.

Unseen families. On five unseen WikiArt families, WEAVE gives the highest CLIP-S and DINO-C and the lowest LPIPS (Table 3); SaMam is 0.015 higher on DINO-S but has higher LPIPS.

Method	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-S $\uparrow$	DINO-C $\uparrow$
Ours (WEAVE)	0.7742	0.2802	0.5313	0.7587
SaMam	0.7577	0.3990	0.5466	0.7369
SaMST	0.7485	0.6213	0.3913	0.2680

Table 3: Zero-shot evaluation on Other5 (750 pairs) using the D5 checkpoint.

Basis and resolution. Daubechies-2 slightly improves CLIP-S but worsens LPIPS; Haar gives the better measured trade-off and has exact orthogonality with compact support. Native-resolution training also outperforms upsampling a low-resolution checkpoint, consistent with style information being carried by high-frequency bands that low-resolution training compresses.

Plug-and-Play Extension to Pre-trained Diffusion Models

We add the same Haar and endpoint AdaIN operation to a frozen SD1.5 image-to-image pipeline without training. The base editor uses an empty prompt, strength 0.5, and 20 DDIM steps; the plugin aligns LH/HL/HH at scale 0.8 while leaving LL unchanged. On 600 off-diagonal D5 pairs, both CLIP-S and DINO-S increase ($p < 10^{-15}$, paired t -test) with little content change, and the improvement remains on the full 750-pair board. The DWT-AdaIN-IDWT operation adds 4.9 ms per image. This experiment tests transferability of the operator, not a claim that the small network and SD1.5 are otherwise equivalent.

Mechanism Analysis

Gradient spectrum. The unweighted LL loss and its head gradient dominate the high-frequency terms; LL remains the largest input-gradient band after down-weighting. Thus the imbalance is not created solely by the chosen weights, although this does not establish a universal cause.

Subband roles. LH/HL carry most measured style change, LL retains structure, and headless HH changes through endpoint AdaIN. Removing AdaIN suppresses LH transfer, while WCT adds content drift for little style gain, supporting the division between learned transport and endpoint alignment.

Discussion

WEAVE separates coarse displacement, organized residual transport, and high-frequency moments. Gradient probes, component removal, and Latent-WCT consistently support frequency dominance as an important cause of the identity shortcut in this system, not a universal explanation for every architecture.

Scope and limitations. IDT and TGT are empirical references, not perceptual bounds. WEAVE is domain-conditional; palette-dominated and multi-scale styles remain harder, and broader checkpoint families are needed to test the mechanism beyond D5, R5, and Other5.

Conclusion

The IDT-TGT sandwich gives style and content metrics an explicit direction: IDT identifies methods stuck near the source, while TGT identifies those that abandon it. WEAVE addresses this conflict in frequency coordinates. A source-aligned target limits unnecessary structure motion, band-weighted flow learns organized transport, and endpoint AdaIN supplies high-frequency statistics without repeatedly perturbing the trajectory.

Experiments across datasets, unseen styles, and complementary metrics support this division. Gradient and subband probes link the identity shortcut to low-frequency dominance, while interventions show that analytic statistics and learned transport are both necessary. The native high-resolution and frozen-SD1.5 results suggest scale and transfer potential. The gate/gradient transition also turns the diagnosis into a practical stopping rule, selecting the style peak without decoded validation. FD-DB frames the same appearance-geometry

tension in synthetic-to-real translation (Zang, Hu, and Song 2026), pointing to structure-preserving latent augmentation.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., “Impressionism”) rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. The D5 run stops after four epochs and takes 82.8 seconds (1.4 minutes) on a single RTX 3060 (12 GB; batch 96), without online image evaluation. Generation-only inference for 750 pairs needs about 106 seconds for the oriented-HF model (8-step ODE + VAE decode), and a full metrics pass adds metric extraction and writing overhead. The model has 1.04M trainable parameters plus the frozen 84M VAE. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

References

- Alibaba Cloud. 2024. Qwen-Image: Advanced Generative Vision Models. <https://qwenlm.github.io/>.
- Black Forest Labs. 2024. FLUX.1: A State-of-the-Art Text-to-Image Model. <https://blackforestlabs.ai/>.
- Cai, M.; Zhang, H.; Huang, H.; Geng, Q.; Li, Y.; and Huang, G. 2021. Frequency Domain Image Translation: More Photo-Realistic, Better Identity-Preserving. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Chung, J.; Hyun, S.; and Heo, J.-P. 2024. Style Injection in Diffusion: A Training-free Approach for Adapting Large-scale Diffusion Models for Style Transfer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.
- Cuturi, M. 2013. Sinkhorn Distances: Lightspeed Computation of Optimal Transport. *Advances in Neural Information Processing Systems*.
- Daubechies, I. 1992. Ten Lectures on Wavelets. *SIAM*.
- De Bortoli, V.; Thornton, J.; Heng, J.; and Doucet, A. 2021. Diffusion Schrödinger Bridge with Applications to Score-Based Generative Modeling. *arXiv preprint arXiv:2106.01357*.
- Deng, Y.; Tang, F.; Dong, W.; Sun, W.; Huang, F.; and Xu, C. 2022. StyTr2: Image Style Transfer with Transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2015. A Neural Algorithm of Artistic Style. In *arXiv preprint arXiv:1508.06576*.
- Gatys, L. A.; Ecker, A. S.; and Bethge, M. 2016. Image Style Transfer Using Convolutional Neural Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Gim, H.; Yoo, J.; and Uh, Y. 2024. Content-Adaptive Style Transfer: A Training-Free Approach with VQ Autoencoders. In *Proceedings of the Asian Conference on Computer Vision*.
- Gu, A.; and Dao, T. 2023. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. *arXiv preprint arXiv:2312.00752*.
- Hertz, A.; Mokady, R.; Tenenbaum, J.; Aberman, K.; Pritch, Y.; and Cohen-Or, D. 2022. Prompt-to-Prompt Image Editing with Cross Attention Control. *arXiv preprint arXiv:2208.01626*.
- Hertz, A.; Voynov, A.; Fruchter, S.; and Cohen-Or, D. 2024. Style Aligned Image Generation via Shared Attention. In *Proceedings of the International Conference on Learning Representations (ICLR)*.
- Hong, K.; Jeon, S.; Yang, H.; Fu, J.; and Byun, H. 2023. AesPA-Net: Aesthetic Pattern-Aware Style Transfer Networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Huang, X.; and Belongie, S. 2017. Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization. In *ICCV*.
- Kim, G.; Kwon, T.; and Ye, J. C. 2023. Diffusion-based Image Translation using Disentangled Style and Content Representation. In *International Conference on Learning Representations*.
- Kwon, J.; Kim, S.; Lin, Y.; Yoo, S.; and Cha, J. 2024. AesFA: An Aesthetic Feature-Aware Arbitrary Neural Style Transfer. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Léonard, C. 2014. A Survey of the Schrödinger Problem and Some of Its Connections with Optimal Transport. *Discrete and Continuous Dynamical Systems - A*.
- Li, Y.; Fang, C.; Yang, J.; Wang, Z.; Lu, X.; and Yang, M.-H. 2017. Universal Style Transfer via Feature Transforms. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Lipman, Y.; Chen, R. T. Q.; Ben-Hamu, H.; Nickel, M.; and Le, M. 2022. Flow Matching for Generative Modeling. *arXiv preprint arXiv:2210.02747*.
- Liu, G.-H.; Vahdat, A.; Huang, D.-A.; Theodorou, E. A.; Nie, W.; and Anandkumar, A. 2023. I2SB: Image-to-Image Schrödinger Bridge. In *International Conference on Machine Learning*.
- Liu, H.; Wang, L.; Guan, W.; Zhang, Y.; and Guo, Y. 2024. Pluggable Style Representation Learning for Multi-Style Transfer. In *Proceedings of the Asian Conference on Computer Vision*.
- Liu, H.; Wang, L.; Zhang, Y.; Yu, Z.; and Guo, Y. 2025. SaMam: Style-aware State Space Model for Arbitrary Image Style Transfer. *arXiv preprint arXiv:2503.15934*.

- Liu, X.; Gong, C.; and Liu, Q. 2023. Flow Straight and Fast: Learning to Generate and Transfer Data with Rectified Flow. In *International Conference on Learning Representations*.
- Lu, Z.-S.; Vukoti, S.; Hadji, I.; Dorken, W.; and Levi, G. 2024. Z-STAR: Training-Free Zero-Shot Style Transfer via Attention Rearrangement. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Meng, C.; He, Y.; Song, Y.; Song, J.; Blackwell, J.; Jia, J.; and Ermon, S. 2021. SDEdit: Guided Image Synthesis and Editing with Stochastic Differential Equations. In *Proceedings of the IEEE International Conference on Computer Vision*.
- Park, D. Y.; and Lee, K. H. 2019. Arbitrary Style Transfer with Style-Attentional Networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Park, T.; Efros, A. A.; Zhang, R.; and Zhu, J.-Y. 2020. Contrastive Learning for Unpaired Image-to-Image Translation. In *Proceedings of the European Conference on Computer Vision*.
- Peyré, G.; and Cuturi, M. 2019. Computational Optimal Transport. *Foundations and Trends in Machine Learning*.
- Phung, H.; Dao, Q.; and Tran, A. 2023. Wavelet Diffusion Models Are Fast and Scalable Image Generators. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Rahaman, N.; Baratin, A.; Arpit, D.; Draxler, F.; Li, M.; Hamprecht, F.; Bengio, Y.; and Courville, A. 2019. On the Spectral Bias of Neural Networks. In *ICML*.
- Rombach, R.; Blattmann, A.; Lorenz, D.; Esser, P.; and Ommer, B. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *CVPR*.
- Shi, Y.; De Bortoli, V.; Campbell, A.; and Doucet, A. 2023. Diffusion Schrödinger Bridge Matching. In *Advances in Neural Information Processing Systems*.
- Su, X.; Song, J.; Meng, C.; and Ermon, S. 2023. Dual Diffusion Implicit Bridges for Image-to-Image Translation. In *International Conference on Learning Representations*.
- Wright, M.; and Ommer, B. 2022. ArtFID: Quantitative Evaluation of Neural Style Transfer. *arXiv preprint arXiv:2207.12280*.
- Xu, Z.-Q. J.; Zhang, Y.; and Luo, T. 2022. Overview frequency principle/spectral bias in deep learning. *arXiv preprint arXiv:2201.07395*.
- Ye, H.; Zhang, J.; Liu, S.; Han, X.; and Yang, W. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. *arXiv preprint arXiv:2308.06721*.
- Yoo, J.; Uh, Y.; Chun, S.; Kang, B.; and Ha, J.-W. 2019. Photorealistic Style Transfer via Wavelet Transforms. In *ICCV*.
- Zang, C.; Hu, J.; and Song, X. 2026. FD-DB: Frequency-Decoupled Dual-Branch Network for Unpaired Synthetic-to-Real Domain Translation.
- Zhang, C.; Xu, X.; Wang, L.; Dai, Z.; and Yang, J. 2024a. S2WAT: Image Style Transfer via Hierarchical Vision Transformer Using Strips Window Attention. In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Zhang, R.; Isola, P.; Efros, A. A.; Shechtman, E.; and Wang, O. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *CVPR*.
- Zhang, T.; and Tang, H. 2025. Style Transfer: A Decade Survey. *arXiv preprint arXiv:2506.19278*.
- Zhang, Y.; Huang, N.; Tang, F.; Huang, H.; Ma, C.; Dong, W.; and Xu, C. 2023. Inversion-Based Style Transfer with Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.
- Zhang, Y.; et al. 2024b. ArtBank: Artistic Style Transfer with Pre-trained Diffusion Model and Implicit Style Prompt. In *Proceedings of the AAAI Conference on Artificial Intelligence*.